



Online

Object Oriented Programming

Computer Systems Engineering

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Activity 2: Analyzing the American Stock Exchange

Module 2: Methods and Standard I/O Streams

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1. Description of the solution

This activity models a stock from the American Stock Exchange with an object-oriented approach. The program is split into two files: the **Stock** class, which holds the data and the behavior of a single stock, and a test program that creates one stock and reports its price change.

The **Stock** class keeps four private data fields: **symbol** and **name** (both *String*), and **previousClosingPrice** and **currentPrice** (both *double*). The constructor receives the symbol and the name when the object is created. The two prices are assigned afterward through **setPreviousPrice** and **setNewPrice**. The method **getChangePercent** applies the formula below, so the result is positive when the price rises and negative when it falls. A getter was added for each field to read the values from outside the class while the fields stay private, which keeps the object properly encapsulated.

```
getChangePercent = ((currentPrice - previousClosingPrice) /  
previousClosingPrice) × 100
```

The test program **TestStock** creates a **Stock** for Netflix with the symbol **NFLX** and the name **Netflix Corporation**. It sets the previous closing price to 337.5 and the current price to 345.23, and then prints the stock data together with the price-change percentage. With those values the change is $((345.23 - 337.5) / 337.5) \times 100 = 2.29\%$, a small rise that the program also states in words.

2. Source code

Stock.java

```

// Models a stock traded on the American Stock Exchange.
public class Stock {
    private String symbol;           // the stock's symbol
    private String name;             // the stock's name
    private double previousClosingPrice; // price for the previous day
    private double currentPrice;      // price for the current time

    // Constructor: creates a stock with the specified symbol and name.
    public Stock(String symbol, String name) {
        this.symbol = symbol;
        this.name = name;
    }

    public String getSymbol() { return symbol; }
    public String getName() { return name; }
    public double getPreviousClosingPrice() { return previousClosingPrice; }
    public double getCurrentPrice() { return currentPrice; }

    // Sets the value of the data field previousClosingPrice.
    public void setPreviousPrice(double previousClosingPrice) {
        this.previousClosingPrice = previousClosingPrice;
    }

    // Sets the value of the data field currentPrice.
    public void setNewPrice(double currentPrice) {
        this.currentPrice = currentPrice;
    }

    // Returns the percentage changed from previousClosingPrice to currentPrice.
    public double getChangePercent() {
        return ((currentPrice - previousClosingPrice) / previousClosingPrice) * 100;
    }
}

```

TestStock.java

```
// Test program for the Stock class.
public class TestStock {
    public static void main(String[] args) {
        // Create a Stock object for Netflix with its symbol and name
        Stock netflix = new Stock("NFLX", "Netflix Corporation");

        // Set the previous closing price and the new current price
        netflix.setPreviousPrice(337.5);
        netflix.setNewPrice(345.23);

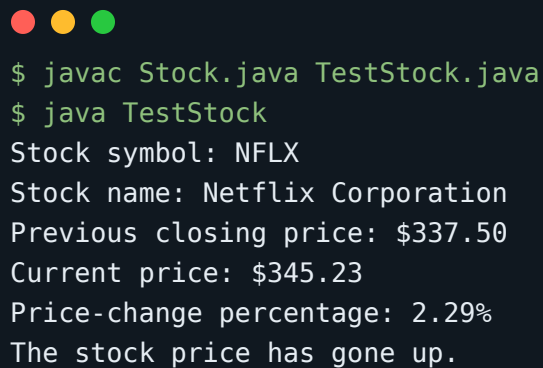
        // Display the stock information
        System.out.println("Stock symbol: " + netflix.getSymbol());
        System.out.println("Stock name: " + netflix.getName());
        System.out.printf("Previous closing price: $%.2f%n",
netflix.getPreviousClosingPrice());
        System.out.printf("Current price: $%.2f%n", netflix.getCurrentPrice());

        // Display the price-change percentage
        double change = netflix.getChangePercent();
        System.out.printf("Price-change percentage: %.2f%%n", change);

        // Indicate whether the stock went up or down
        if (change > 0) {
            System.out.println("The stock price has gone up.");
        } else if (change < 0) {
            System.out.println("The stock price has gone down.");
        } else {
            System.out.println("The stock price has not changed.");
        }
    }
}
```

3. Execution evidence

The two files compile with `javac` without errors or warnings, and the test program produces the expected output. The screenshot below shows the compilation and the run:



```
$ javac Stock.java TestStock.java
$ java TestStock
Stock symbol: NFLX
Stock name: Netflix Corporation
Previous closing price: $337.50
Current price: $345.23
Price-change percentage: 2.29%
The stock price has gone up.
```

The current price (345.23) is higher than the previous closing price (337.5), so the change is positive: the Netflix stock went up 2.29 % from one day to the next.